

I

Python script to build and verify Percentage questions

gen_profitloss.py

```
import json
```

```
Q = []
```

```
def add(text, options, correct_idx, explain):
```

```
    assert len(options) == 4
```

```
    Q.append({"q": text, "options": options, "answer": correct_idx, "exp": explain})
```

```
# 1
```

```
add("A man buys an article for ₹450 and sells it for ₹540. Find his profit percentage.",  
    ["20%", "18%", "22%", "15%"], 0,  
    "Profit = 540-450 = 90. Profit% = 90/450×100 = 20%.")
```

```
# 2
```

```
add("A shopkeeper sells an article at a loss of 12%. If he had sold it for ₹99 more, he  
would have gained 10%. Find the cost price.",  
    ["₹450", "₹500", "₹420", "₹480"], 0,  
    "Let CP = x.  $1.10x - 0.88x = 99 \rightarrow 0.22x = 99 \rightarrow x = 450$ .")
```

```
# 3
```

```
add("The cost price of 20 articles is equal to the selling price of 16 articles. Find the profit  
percentage.",  
    ["25%", "20%", "22%", "16%"], 0,  
    "CP of 1 article=x, so CP of 20 = 20x = SP of 16  $\rightarrow$  SP per article =  $20x/16 = 1.25x$ . Profit%  
= 25%.")
```

```
# 4
```

```
add("A trader marks his goods 40% above cost price and allows a discount of 25%. Find  
his profit percentage.",  
    ["5%", "10%", "8%", "15%"], 0,  
    "CP=100, MP=140, SP=140×0.75=105. Profit% = 5%.")
```

```
# 5
```

```
add("By selling a table for ₹1140, a man loses 5%. At what price should he sell it to gain  
8%?",  
    ["₹1296", "₹1300", "₹1250", "₹1280"], 0,  
    "CP =  $1140/0.95 = 1200$ . SP for 8% gain =  $1200 \times 1.08 = 1296$ .")
```

6

add("A dishonest dealer professes to sell his goods at cost price but uses a weight of 800 gm instead of 1 kg. Find his profit percentage.",

["25%", "20%", "22%", "18%"], 0,

"He gives 800g but charges for 1000g. Profit% = $(200/800) \times 100 = 25\%$."

7

add("A man sells two articles, each for ₹1200. On one he gains 20% and on the other he loses 20%. Find his overall profit or loss percentage.",

["4% loss", "No profit no loss", "4% profit", "2% loss"], 0,

"When SP is equal and gain%=loss%, there is always an overall loss of $(\text{common}\%/10)^2 = (20/10)^2 = 4\%$."

8

add("A shopkeeper allows a discount of 10% on the marked price of an article and still makes a profit of 26%. If the article costs him ₹450, find the marked price.",

["₹630", "₹600", "₹650", "₹610"], 0,

"SP = $450 \times 1.26 = 567$. MP $\times 0.90 = 567 \rightarrow \text{MP} = 567/0.9 = 630$."

9

add("A man buys a watch for ₹800 and sells it at a loss of 15%. He then buys it back at a price 10% less than what he sold it for, and sells it again at a profit of 20%. Find his overall profit or loss.",

["₹34.80 profit", "₹40 profit", "₹30 loss", "₹36 profit"], 0,

"SP1 = $800 \times 0.85 = 680$. Buys back at $680 \times 0.90 = 612$. Sells at $612 \times 1.20 = 734.40$. Net: spent 800, got 680, spent 612, got 734.40. Net cash flow = $-800 + 680 - 612 + 734.40 = 2.40$? Let's recompute: overall profit = $(680 - 800) + (734.40 - 612) = -120 + 122.40 = 2.40$. This needs revision."

fix 9 - recompute properly and simplify problem to avoid ambiguity

Q.pop()

add("A man buys a watch for ₹800 and sells it at a loss of 15%. He later buys an identical watch for ₹680 and sells it at a profit of 20%. Find his overall profit or loss on the two transactions combined.",

["₹16 profit", "₹20 loss", "₹10 profit", "No profit no loss"], 0,

"Transaction 1: CP=800, SP= $800 \times 0.85 = 680$, loss=120. Transaction 2: CP=680, SP= $680 \times 1.20 = 816$, profit=136. Net = $136 - 120 = ₹16$ profit."

10

add("A retailer buys 30 pens at the marked price of 25 pens from a wholesaler. If he sells these pens giving a discount of 5%, what is his profit percentage?",

["14%", "12%", "16%", "10%"], 0,

"Let marked price of 1 pen = ₹1. CP of 30 pens = price of 25 = ₹25. SP of 30 pens at 5% discount = $30 \times 0.95 = ₹28.5$. Profit% = $(28.5 - 25) / 25 \times 100 = 14\%$."

11

add("A man sold an article at a loss of 15%. Had he sold it for ₹300 more, he would have gained 5%. Find the selling price of the article to gain 10%.")

Input validation errors occurred: description: Field required

Input validation errors occurred: description: Field required

Complete the profit-loss question generator with remaining questions 11-25

Successfully replaced string in /home/claude/gen_profitloss.py

Good, that's clean now (25 questions, question 9 properly fixed via pop/re-add).
Let me run it.

Now Ratio & Proportion (25 questions):

Python script to build Ratio and Proportion questions

Now Number System (25 questions):

Python script to build and verify Number System questions

Python script to build Data Interpretation question sets with computed answers

Python script to build coding questions dataset (Java hints+solutions)

Now I'll build the full exam HTML file. Given the scale (125 MCQs + 16 coding problems), I'll assemble it via a Python script for reliability.

`build_html.py`

```
import json
```

```
perc = json.load(open('percentage.json'))
pl = json.load(open('profitloss.json'))
ratio = json.load(open('ratio.json'))
ns = json.load(open('numsys.json'))
di = json.load(open('di.json'))
coding = json.load(open('coding.json'))
```

```
DATA = {
    "aptitude": {
        "percentage": {"label": "Percentage", "questions": perc},
```

```

    "profitloss": {"label": "Profit & Loss", "questions": pl},
    "ratio": {"label": "Ratio & Proportion", "questions": ratio},
    "numsys": {"label": "Number System", "questions": ns},
    "di": {"label": "Data Interpretation", "sets": di},
  },
  "coding": coding
}

```

```

data_json = json.dumps(DATA, ensure_ascii=False)
data_json = data_json.replace("</script", "<\\script").replace("<script", "<\\script")

```

```

print("Data JSON length:", len(data_json))

```

```

with open("data_blob.js", "w") as f:
    f.write("const EXAM_DATA = " + data_json + ";\n")

```

```

print("done")

```

style.css

```

:root{
  --bg:#f3f5f8;
  --panel:#ffffff;
  --border:#e2e6ec;
  --text:#1f2733;
  --muted:#6b7583;
  --blue:#0b5fce;
  --blue-dark:#0847a3;
  --blue-tint:#eaf1fd;
  --green:#1a9c5a;
  --green-tint:#e7f7ee;
  --red:#d5382f;
  --red-tint:#fdecea;
  --amber:#b8860b;
  --grey-tint:#eef0f3;
  --radius:8px;
  --mono: "SFMono-Regular", Consolas, "Liberation Mono", Menlo, monospace;
  --sans: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Helvetica, Arial, sans-serif;
}
*{box-sizing:border-box;}
html,body{height:100%;}
body{

```

```

margin:0;
font-family:var(--sans);
background:var(--bg);
color:var(--text);
font-size:14px;
line-height:1.5;
}
button{font-family:inherit;cursor:pointer;}
::selection{background:#bcd6fb;}

/* ----- Top bar ----- */
.topbar{
  position:sticky;top:0;z-index:50;
  display:flex;align-items:center;gap:16px;
  background:var(--panel);
  border-bottom:1px solid var(--border);
  padding:10px 20px;
}
.brand{display:flex;align-items:center;gap:10px;}
.brand-mark{
  width:30px;height:30px;border-radius:7px;
  background:linear-gradient(135deg,var(--blue),#3b82f6);
  display:flex;align-items:center;justify-content:center;
  color:#fff;font-weight:700;font-size:13px;
}
.brand-text{display:flex;flex-direction:column;line-height:1.15;}
.brand-text b{font-size:14.5px;letter-spacing:.2px;}
.brand-text span{font-size:11px;color:var(--muted);}

.section-tabs{display:flex;gap:4px;margin-left:8px;}
.section-tab{
  padding:8px 16px;border-radius:7px;border:1px solid transparent;
  background:transparent;color:var(--muted);font-weight:600;font-size:13px;
}
.section-tab.active{background:var(--blue-tint);color:var(--blue-dark);}
.section-tab:hover:not(.active){background:var(--grey-tint);}

.topbar-spacer{flex:1;}

.timer{
  display:flex;align-items:center;gap:6px;
  font-variant-numeric:tabular-nums;

```

```

    font-weight:600;color:var(--text);
    background:var(--grey-tint);
    padding:6px 12px;border-radius:20px;font-size:13px;
}
.timer svg{width:14px;height:14px;color:var(--muted);}

.btn{
    border:1px solid var(--border);background:var(--panel);color:var(--text);
    padding:8px 14px;border-radius:7px;font-weight:600;font-size:13px;
}
.btn:hover{background:var(--grey-tint);}
.btn-primary{background:var(--blue);border-color:var(--blue);color:#fff;}
.btn-primary:hover{background:var(--blue-dark);}
.btn-danger{background:#fff;border-color:var(--red);color:var(--red);}
.btn-danger:hover{background:var(--red-tint);}
.btn-ghost{border-color:transparent;background:transparent;}
.btn-ghost:hover{background:var(--grey-tint);}
.btn.disabled{opacity:.45;cursor:not-allowed;}
.btn-sm{padding:5px 10px;font-size:12px;border-radius:6px;}

/* ----- Layout ----- */
.app{display:none;max-width:1300px;margin:0 auto;padding:18px 20px 60px;}
.app.active{display:block;}

.subtabs{display:flex;flex-wrap:wrap;gap:6px;margin-bottom:16px;}
.subtab{
    padding:7px 14px;border-radius:20px;border:1px solid var(--border);
    background:var(--panel);color:var(--muted);font-weight:600;font-size:12.5px;
}
.subtab.active{background:var(--blue);border-color:var(--blue);color:#fff;}
.subtab .count{opacity:.75;font-weight:500;margin-left:4px;}

.layout{display:grid;grid-template-columns:1fr 300px;gap:18px;align-items:start;}
@media (max-width:880px){.layout{grid-template-columns:1fr;}}

.panel{
    background:var(--panel);border:1px solid var(--border);border-radius:var(--radius);
    padding:22px 24px;
}

/* ----- Question ----- */
.q-meta{display:flex;align-items:center;justify-content:space-between;margin-

```

```

bottom:14px;}
.q-tag{
  font-size:11.5px;font-weight:700;color:var(--blue-dark);background:var(--blue-tint);
  padding:3px 9px;border-radius:5px;letter-spacing:.3px;text-transform:uppercase;
}
.q-index{color:var(--muted);font-size:12.5px;font-weight:600;}
.q-text{font-size:15.5px;font-weight:600;margin:0 0 18px;color:var(--text);line-
height:1.55;}

.options{display:flex;flex-direction:column;gap:9px;margin-bottom:22px;}
.option{
  display:flex;align-items:flex-start;gap:11px;
  border:1px solid var(--border);border-radius:7px;padding:11px 14px;
  background:#fff;
}
.option:hover{border-color:#c4d5f2;background:#fafcff;}
.option.selected{border-color:var(--blue);background:var(--blue-tint);}
.option input{margin-top:3px;accent-color:var(--blue);}
.option label{flex:1;cursor:pointer;font-size:14px;}
.option .opt-letter{
  width:20px;height:20px;border-radius:50%;background:var(--grey-tint);
  display:flex;align-items:center;justify-content:center;font-size:11px;font-weight:700;
  color:var(--muted);flex-shrink:0;
}
.option.selected .opt-letter{background:var(--blue);color:#fff;}

/* review states after submit */
.option.correct-ans{border-color:var(--green);background:var(--green-tint);}
.option.correct-ans .opt-letter{background:var(--green);color:#fff;}
.option.wrong-ans{border-color:var(--red);background:var(--red-tint);}
.option.wrong-ans .opt-letter{background:var(--red);color:#fff;}
.explain{
  margin-top:14px;padding:12px 14px;background:var(--grey-tint);border-radius:7px;
  font-size:13px;color:var(--text);display:none;
}
.explain.show{display:block;}
.explain b{color:var(--blue-dark);}

.q-nav-row{display:flex;justify-content:space-between;align-
items:center;gap:10px;padding-top:6px;border-top:1px solid var(--border);margin-
top:4px;}
.q-nav-row .left-btns,.q-nav-row .right-btns{display:flex;gap:8px;}

```

```

/* ----- Palette sidebar ----- */
.side-title{font-size:12px;font-weight:700;color:var(--muted);text-
transform:uppercase;letter-spacing:.4px;margin:0 0 10px;}
.legend{display:flex;flex-wrap:wrap;gap:10px;margin-bottom:14px;font-
size:11.5px;color:var(--muted);}
.legend span{display:flex;align-items:center;gap:5px;}
.dot{width:9px;height:9px;border-radius:50%;display:inline-block;}
.dot.answered{background:var(--green);}
.dot.notanswered{background:var(--red);}
.dot.notvisited{background:#c7ccd4;}
.dot.current{background:var(--blue);}

.palette{display:grid;grid-template-columns:repeat(5,1fr);gap:7px;margin-
bottom:16px;}
.pnum{
  aspect-ratio:1;border-radius:6px;border:1px solid var(--border);
  background:#fff;font-weight:700;font-size:12.5px;color:var(--muted);
  display:flex;align-items:center;justify-content:center;
}
.pnum.answered{background:var(--green);border-color:var(--green);color:#fff;}
.pnum.notanswered{background:var(--red-tint);border-color:var(--red);color:var(--
red);}
.pnum.current{outline:2px solid var(--blue);outline-offset:1px;}
.pnum:hover{border-color:var(--blue);}

.topic-summary{border-top:1px solid var(--border);padding-top:14px;margin-
top:4px;font-size:12.5px;color:var(--muted);}
.topic-summary div{display:flex;justify-content:space-between;padding:3px 0;}
.topic-summary b{color:var(--text);}

/* ----- Data interpretation chart ----- */
.di-chart-box{
  border:1px solid var(--border);border-radius:8px;padding:18px 20px;margin-
bottom:18px;background:#fbfcfe;
}
.di-chart-title{font-weight:700;font-size:13.5px;margin-bottom:14px;color:var(--text);}
.bar-chart{display:flex;align-items:flex-end;gap:18px;height:180px;padding:0
6px;border-bottom:1px solid var(--border);}
.bar-group{display:flex;flex-direction:column;align-items:center;justify-content:flex-
end;flex:1;height:100%;gap:3px;}
.bar-set{display:flex;gap:3px;align-items:flex-end;height:100%;}

```

```

.bar{width:16px;border-radius:3px 3px 0 0;position:relative;}
.bar .bar-val{position:absolute;top:-16px;left:50%;transform:translateX(-50%);font-size:10px;font-weight:700;color:var(--muted);white-space:nowrap;}
.bar-label{font-size:11px;color:var(--muted);margin-top:8px;font-weight:600;}
.chart-legend{display:flex;gap:16px;margin-top:12px;font-size:11.5px;color:var(--muted);}
.chart-legend span{display:flex;align-items:center;gap:5px;}
.legend-swatch{width:10px;height:10px;border-radius:3px;}

.pie-wrap{display:flex;gap:26px;align-items:center;flex-wrap:wrap;}
.pie{width:170px;height:170px;border-radius:50%;flex-shrink:0;}
.pie-legend{display:flex;flex-direction:column;gap:7px;font-size:12.5px;}
.pie-legend .row{display:flex;align-items:center;gap:8px;}

.di-table{width:100%;border-collapse:collapse;font-size:12.5px;}
.di-table th,.di-table td{border:1px solid var(--border);padding:7px 10px;text-align:center;}
.di-table th{background:var(--grey-tint);font-weight:700;color:var(--text);}
.di-table td:first-child,.di-table th:first-child{text-align:left;font-weight:600;}

/* ----- Coding section ----- */
.code-layout{display:grid;grid-template-columns:280px 1fr;gap:18px;align-items:start;}
@media (max-width:880px){.code-layout{grid-template-columns:1fr;}}
.code-list{display:flex;flex-direction:column;gap:6px;}
.code-item{
  display:flex;align-items:center;gap:10px;padding:10px 12px;border-radius:7px;
  border:1px solid var(--border);background:#fff;text-align:left;
}
.code-item:hover{border-color:#c4d5f2;}
.code-item.active{border-color:var(--blue);background:var(--blue-tint);}
.code-item .ci-num{
  width:22px;height:22px;border-radius:50%;background:var(--grey-tint);color:var(--muted);
  display:flex;align-items:center;justify-content:center;font-size:11px;font-weight:700;flex-shrink:0;
}
.code-item.solved .ci-num{background:var(--green);color:#fff;}
.code-item.hints .ci-num{background:var(--amber);color:#fff;}
.code-item.viewed .ci-num{background:#c7ccd4;color:#fff;}
.ci-body{flex:1;min-width:0;}
.ci-title{font-size:12.5px;font-weight:600;color:var(--text);white-space:nowrap;overflow:hidden;text-overflow:ellipsis;}

```

```

.diff-badge{font-size:10px;font-weight:700;padding:2px 7px;border-radius:10px;text-
transform:uppercase;letter-spacing:.3px;}
.diff-Easy{background:var(--green-tint);color:var(--green);}
.diff-Medium{background:#fff4e0;color:#b8860b;}
.diff-Hard{background:var(--red-tint);color:var(--red);}

.code-title-row{display:flex;align-items:center;gap:10px;margin-bottom:8px;}
.code-title-row h2{font-size:17px;margin:0;}

.io-block{
  background:#0f1720;color:#dbe4ee;border-radius:7px;padding:14px 16px;
  font-family:var(--mono);font-size:12.5px;white-space:pre-wrap;margin:14px 0
18px;line-height:1.6;
}
.io-block .lbl{color:#fb2ff;font-weight:700;}

.hints-block{display:flex;flex-direction:column;gap:10px;margin-bottom:18px;}
.hint-btn-row{display:flex;gap:8px;flex-wrap:wrap;}
.hint-card{
  display:none;background:var(--blue-tint);border:1px solid #cfe0fa;border-radius:7px;
  padding:11px 14px;font-size:13px;color:var(--text);
}
.hint-card.show{display:block;}
.hint-card b{color:var(--blue-dark);}

.solution-box{margin-top:8px;}
.code-block{
  background:#0f1720;color:#e3eaf2;border-radius:7px;padding:16px 18px;
  font-family:var(--mono);font-size:12.5px;line-height:1.65;white-space:pre;overflow-
x:auto;display:none;
}
.code-block.show{display:block;}

.status-row{display:flex;gap:8px;flex-wrap:wrap;margin-top:18px;padding-
top:16px;border-top:1px solid var(--border);}
.status-btn{padding:8px 14px;border-radius:7px;border:1px solid var(--
border);background:#fff;font-size:12.5px;font-weight:600;color:var(--muted);}
.status-btn.sel-solved.active{background:var(--green);border-color:var(--
green);color:#fff;}
.status-btn.sel-hints.active{background:var(--amber);border-color:var(--
amber);color:#fff;}
.status-btn.sel-viewed.active{background:#8a92a0;border-color:#8a92a0;color:#fff;}

```

```

.code-progress{font-size:12.5px;color:var(--muted);margin-bottom:12px;}

/* ----- Results screen ----- */
.results-overlay{
  position:fixed;inset:0;background:rgba(15,20,28,.55);z-index:200;
  display:none;align-items:flex-start;justify-content:center;overflow:auto;padding:40px
  16px;
}
.results-overlay.show{display:flex;}
.results-card{background:#fff;border-radius:12px;max-
width:760px;width:100%;padding:32px 34px 30px;}
.results-card h2{margin:0 0 4px;font-size:20px;}
.results-card .sub{color:var(--muted);font-size:13px;margin-bottom:22px;}
.score-hero{
  display:flex;align-items:center;gap:22px;background:var(--blue-tint);border-
  radius:10px;
  padding:20px 24px;margin-bottom:22px;
}
.score-big{font-size:34px;font-weight:800;color:var(--blue-dark);}
.score-big span{font-size:16px;font-weight:600;color:var(--muted);}
.score-pct{font-size:13px;color:var(--muted);}
.results-table{width:100%;border-collapse:collapse;margin-bottom:18px;font-
size:13px;}
.results-table th,.results-table td{padding:9px 10px;border-bottom:1px solid var(--
border);text-align:left;}
.results-table th{color:var(--muted);font-weight:700;text-transform:uppercase;font-
size:10.5px;letter-spacing:.3px;}
.results-table td.num{text-align:center;}
.results-actions{display:flex;gap:10px;justify-content:flex-end;margin-top:10px;}

.badge-note{font-size:11.5px;color:var(--muted);background:var(--grey-
tint);padding:10px 12px;border-radius:7px;margin-bottom:16px;}

@media(max-width:600px){
  .topbar{flex-wrap:wrap;}
  .section-tabs{order:3;width:100%;}
  .palette{grid-template-columns:repeat(5,1fr);}
  .results-card{padding:22px 18px;}
}

```

Main application JavaScript for the exam interface

shell.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>TCS Mock Test — Programming & Numerical Ability</title>
<style>
/*__CSS__*/
</style>
</head>
<body>

<div class="topbar">
  <div class="brand">
    <div class="brand-mark">TX</div>
    <div class="brand-text">
      <b>TCS-Style Mock Test</b>
      <span>Programming & Numerical Ability</span>
    </div>
  </div>
  <div class="section-tabs">
    <button class="section-tab active" data-section="aptitude">Aptitude (125
MCQs)</button>
    <button class="section-tab" data-section="coding">Programming (16
problems)</button>
  </div>
  <div class="topbar-spacer"></div>
  <div class="timer">
    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><circle
cx="12" cy="12" r="9"/><path d="M12 7v5l3 3"/></svg>
    <span id="timerText">00:00:00</span>
  </div>
  <button class="btn btn-primary" id="btnSubmit">Submit Exam</button>
</div>

<!-- ===== APTITUDE ===== -->
<div class="app active" id="appAptitude">
  <div class="subtabs" id="aptSubtabs"></div>
  <div class="layout">
    <div class="panel" id="questionPanel"></div>
    <div class="panel">
```

```

    <div class="side-title">Question Palette</div>
    <div class="legend">
        <span><i class="dot answered"></i>Answered</span>
        <span><i class="dot notanswered"></i>Visited</span>
        <span><i class="dot notvisited"></i>Not visited</span>
    </div>
    <div class="palette" id="palette"></div>
    <div class="side-title">This Topic</div>
    <div class="topic-summary" id="topicSummary"></div>
</div>
</div>
</div>

<!-- ===== CODING ===== -->
<div class="app" id="appCoding">
    <div class="code-progress" id="codeProgress"></div>
    <div class="code-layout">
        <div class="panel">
            <div class="side-title" style="margin-bottom:12px;">Problems</div>
            <div class="code-list" id="codeList"></div>
        </div>
        <div class="panel" id="codePanel"></div>
    </div>
</div>

<!-- ===== RESULTS OVERLAY ===== -->
<div class="results-overlay" id="resultsOverlay">
    <div class="results-card">
        <h2>Your Score</h2>
        <div class="sub">Here's how you did across the mock test.</div>
        <div id="resultsBody"></div>
    </div>
</div>

<script>
/*__DATA__*/
</script>
<script>
/*__APP__*/
</script>
</body>
</html>

```